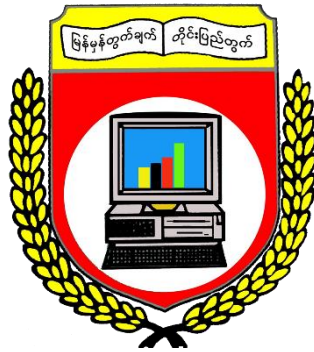


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Project Report On

ANIME INDUSTRY ANALYTICS SYSTEM

Semester IX

DATA AND KNOWLEDGE MINING (IS-212) PROJECT

Submitted By

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TABLE OF CONTENTS

ACKNOWLEDGEMENT

CHAPTER 1	5
INTRODUCTION	5
1.1 Project Background	5
1.2 Objectives of the project	6
1.3 Scope of the project	6
1.4 Overview of the Methodology	7
1.5 Benefits of the project	8
CHAPTER 2	9
2.1 Dataset and Data Statistics	9
2.2 Data Preprocessing	10
2.2.1 Handling Missing Values	10
2.2.2 Data Transformation	12
2.2.3 Feature Selection and Engineering	12
2.3 Data Visualization	14
2.3.1 Numeric Feature Distribution	14
2.3.2 Average Episode and Runtimes by Media Type	16
2.3.3 Genre Trend Over Time	17
CHAPTER 3	18
APPLIED METHODOLOGIES AND MINING	18
3.1 Descriptive Mining	18
3.1.1 Association Rule Mining	18
3.1.2 Producer-Studio Collaboration Patterns	21
3.1.3 Synopsis Text Mining and Narrative Clustering	21
3.2 Predictive Mining	23
3.2.1 Content-Based Hybrid Anime Recommendation	23
3.2.2 Anime Success Classification	23
3.2.3 Genre Trend and Audience-Demand Forecasting	26
CHAPTER 4	28
EVALUATION	28
4.1 Anime Recommendation System Evaluation	28
4.2 Anime Success Classification Performance Evaluation	29
4.2.1 Comparison of Experimental Results	29

4.2.2 ROC and AUC	30
4.2.3 Time-Series Cross-Validation and Threshold Optimization.....	30
4.2.4 Final XGBoost Evaluation and Confusion Matrix.....	31
4.2.5 Findings.....	32
4.3 Forecasting Evaluation.....	32
4.3.1 Genre Release-Volume Forecasting	33
4.3.2 Genre Audience-Demand Forecasting.....	33
CHAPTER 5	35
CONCLUSION.....	35
5.1 Advantages of using this project.....	35
5.2 Disadvantages of using this project/ Limitation	36
REFERENCE	

LIST OF FIGURES

Figure 2.1 Distribution of score and rank.....	14
Figure 2.2 Distribution of popularity.....	15
Figure 2.3 Distribution of Engagement signals.....	15
Figure 2.4 Average number of episodes by anime type.....	16
Figure 2.5 Average number of durations by anime type.....	16
Figure 2.6 Top-10 genre release-count trend, 2010–2025.....	17
Figure 3.1 Genre ↔ Theme Co-occurrenc.....	18
Figure 3.2 Top producer + genre/themes for the Balanced_High outcome.....	20
Figure 3.3 Top studios + genre/themes for the Balanced_High outcome.....	20
Figure 3.4 Silhouette score across candidate cluster counts $k = 8-40$	22
Figure 3.5 Genre audience-demand forecast.....	26
Figure 3.6 Genre release-volume forecast.....	27
Figure 4.1 ROC- Class-weighted Models (Test Set).....	30
Figure 4.2 XGBoost - Test Confusion Matrix.....	32

LIST OF TABLES

Table 2.1 Data dictionary for key attributes used throughout the analysis.....	9
Table 2.2 Composition of the engineered balanced_score metric.....	13
Table 2.3 High raw score but low reach: Balanced Score Reduction.....	13
Table 2.4 Low-to-mid raw score, very high reach: Balanced Score Increase.....	14
Table 3.1 2026–2028 genre audience-demand forecast.....	26
Table 3.2 2026–2028 genre release-volume forecast.....	27
Table 4.1 Performance of Models under Class Weighting.....	29
Table 4.2 Performance of Models under SMOTE.....	29
Table 4.3. Expanding-Window Time-Series Cross-Validation Results.....	31
Table 4.4 Validation Threshold Optimization.....	31
Table 4.5. Final XGBoost Performance.....	31
Table 4.6 Backtest Results for Genre Release-Volume Forecasting.....	33
Table 4.7 Backtest Results for Genre Audience-Demand Forecasting.....	33

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I would also like to mention my thanks to MyAnimeList [2] and the Jikan API [1] for making structured anime metadata available to the public, since this was what enabled the project to take place. I chose anime as the topic of this project since I have enjoyed watching anime from when I was a child. It was a pleasure to be able to study something that I actually like. While working on this project, I also became interested in the amount of information and insight that can be obtained from anime data.

Lastly, I would like to thank all the people who supported me during the development of this project.

CHAPTER 1

INTRODUCTION

Anime has become a rapidly growing and competitive sector of the entertainment industry, producing many thousands of series and movies which broadcast to audiences all over the world. The popularity and success of an anime highly depend on various aspects like storylines, theme, animation style, target audience, and more.

In this project, historical anime data are investigated to identify industry patterns and factors that correlate to audience performance. The project uses data retrieved from the Jikan API [1] to explore different dimensions of anime, such as studios, genres, themes, audience engagement, and performance.

1.1 Project Background

Funding for anime productions comes from many places, and a single studio rarely finances a production. Anime productions receive funding from a “production committee” (Seisaku linkai) system. Production committees are made of manga publishers, broadcasting networks, streaming services, record labels, and merchandise sellers. Each of these companies makes a funding contribution to the anime production. In return, they are allocated commercial rights of their respective goods and services. This system means many companies share the cost of producing an anime while having the opportunity to earn money from many different markets.

The profitability of an anime is not defined by the initial release. Profits may come from many different sources including streaming, licensing, distribution to other countries, showings in theaters, merchandising, and even songs and intellectual property. Of course, much of the budgeting decisions for an anime are made before it is released. So, the real audience response and the profitability of the anime are not known before the anime is done being released. This makes it difficult for production committees to gauge how well a potential investment will be.

Because of these reasons, it is helpful to have historical audience data to better define audience trends and help predict industry potential. MyAnimeList [2] has an archive of a lot of information on many anime titles. This information includes studios, producers, released date, score, popularity, how many times an anime was favorited,

and other audience engagement and focus. The anime data collected using Jikan API [1] in this project is intended to help in finding relationship patterns. The intent behind these findings is to promote a data-driven approach to decision-making with regard to factors correlating with higher audience engagement and greater performance that can guide decisions for new production options, content characteristics and market opportunities.

1.2 Objectives of the project

The primary objectives of this project are:

- ❖ To identify statistically co-occurring combinations of studios, genres, and themes, and pairings of producer partnerships which result in better balance success, using association rules and studios/producers performance aggregation, to guide decisions on studio-matching and content.
- ❖ To identify underlying narrative patterns in anime synopses using TF-IDF, TruncatedSVD, and K-Means clustering, and to examine whether these text-based patterns provide information beyond MAL's predefined genre tags.
- ❖ To build a content-based recommender system that can identify narratively and thematically similar titles, supporting both audience discovery and competitive/comparable-title analysis for a prospective production.
- ❖ To develop a pre-release machine learning classifier to predict whether a new anime concept is likely to reach top-quartile balanced success using only information that is knowable before.
- ❖ To forecast near-term genre-level release-volume and audience demand using time-series forecasting models.

1.3 Scope of the project

The scope of this project entails analyzing historical anime data scraped using Jikan API [1] on MyAnimeList [2]. From this data, insights into different types of anime released between 2010–2025 will be analyzed. Specifically, project boundaries span in terms of the following:

- ❖ Anime Types: The analysis is limited to "TV", "Movie", "ONA" and "OVA" types

- ❖ **Descriptive Analysis:** The descriptive part of the mining uses association rule mining algorithm (FP-Growth) to extract relationship. Text mining was applied to obtain and process significant patterns from anime synopses.
 - ❖ **Predictive Analysis:** The predictive part contains a content-based hybrid recommender to discover similar anime shows, 4 classification models to predict the success of anime release before its production and time-series model for predicting audience demand and release volume over time to understand the pattern on genre release.
 - ❖ **Deployment:** Developed as a web-based interface that presents anime information, images, trailers, statistical analysis, association rules, recommendations, success predictions, and forecasting results in an interactive and user-friendly format.
- Explicit adult content has been excluded because it is unrelated to the mainstream industry investment framework adopted by this project.

1.4 Overview of the Methodology

The project follows a structured data mining workflow:

- ❖ **Data Collection:** Anime data (studio, producers, genres, themes, release date, popularity metrics, images, trailers) is obtained from MyAnimeList [2] via the Jikan REST API [1].
- ❖ **Data Processing:** Clean data and impute missing values, parse list-type fields, standardize entity names and prepare synopsis text for NLP tasks.
- ❖ **Feature Engineering:** Create an overall rating out of the four engagement metrics (score, members, favorites, popularity rank) using weighted percentile metrics.
- ❖ **Pattern Mining and Narrative analysis:** FP-Growth is applied to identify frequent relationships among studios, producers, genres, and themes. For synopsis text mining, TF-IDF is used to represent text [5], followed by TruncatedSVD for dimensionality reduction and K-Means clustering to identify underlying narrative patterns [4].
- ❖ **Predictive Modeling:** Build classifiers (XGBoost [6], Random Forest [7], Decision Tree, Logistic Regression) to determine if an anime can be considered "top-quartile" in success given pre-release or post-release information and do model comparison using time-series cross validation.

- ❖ **Recommendation:** Build content-based hybrid recommender systems that recommend based on synopsis, genre and theme similarity. Since these do not have any ground-truth recommender labels, model performance is compared based on similarity-based metrics.
- ❖ **Forecasting:** Holt Winters Exponential Smoothing [9] is used on historic aggregated releases and customer demands per genre to forecast popularity trend of each genre. Forecasting models are evaluated by MAE, RMSE and Mean Absolute Percentage Error MAPE.
- ❖ The trained models and pre-processing components are then saved as .pkl files post the modeling. These can later be used in web-based application to provide predictions and recommendations without retraining the models.

1.5 Benefits of the project

The following would be benefits associated with this project

- ❖ **Data-Driven Decision Support:** By analyzing historical distribution patterns, the project provides quantitative indication of the relationship between studios, producers, genres and themes and the success of an anime.
- ❖ **Pattern Discovery Beyond Tags:** Use text mining to uncover the potential presence of narrative patterns within anime synopses and analyse correlation with pre-assigned MyAnime List genres.
- ❖ **Pre-Release Decision Support:** Predictive models can be used to evaluate the potential success based on pre-release data and forecasting can be used to predict the upcoming genre trends and audience responses.
- ❖ **Educational Value:** Demonstrates how a complete data mining life cycle from data acquisition, analysis, machine learning, recommendation, forecasting to web deployment can be applied to a real-world entertainment dataset.

CHAPTER 2

DATA PREPARATION AND PREPROCESSING

This chapter discusses the dataset used in this project along with how the data was preprocessed before descriptive mining and predictive modeling. It describes the statistics of the extracted animation records, along with the pre-processing and filtering done and gives detailed descriptions of the pre-processing done on the raw metadata which included cleaning, transforming and engineering.

2.1 Dataset and Data Statistics

The data for this project comes from the Jikan API [1]. This is a public, community-maintained, unofficial REST API that wraps MyAnimeList webpage. The API provides structured anime metadata including titles, genres, themes, studios, producers, source material, airing dates, scores, community engagement counts, images, and trailer links — without requiring authenticated access to MAL itself.

Preliminary data extraction yielded a total of 19,547 raw animation records covering 33 attributes. To ensure that the analysis focused on commercial works that were complete in both scope and narrative, the raw dataset was filtered to retain only core release formats (TV, film, ONA, and OVA). The data was further refined by excluding explicit adult content, filtering out short promotional clips based on duration thresholds for specific formats, and removing entries lacking valid ratings, vote counts, or synopsis. This multi-stage filtering process ultimately produced a consistent final working dataset comprising 8,227 records, which was used throughout all descriptive analysis and predictive modeling phases of this project.

Data Dictionary — Key Attributes

Attribute	Meaning
title / title_english / title_japanese / title_synonym	Romanized, English, Japanese, and alternative titles for the same anime.
type	Delivery format: TV, Movie, ONA, or OVA (after filtering).
source	Origin material: Manga, Light novel, Original, Game, Web manga, Visual novel, Novel, Other/Unknown, etc.

Attribute	Meaning
genres / themes / demographics	MAL-assigned classification tags; demographics indicates an intended audience segment such as Shounen, Shoujo, Seinen, or Josei.
studios / producers	The animation studio(s) that produced the anime and the production-committee member companies.
episodes / duration_minutes	Number of episodes and per-episode (or per-movie) runtime in minutes,
score / scored_by / rank	MAL's official weighted community score, the number of users who submitted a score, and the rank derived from that score.
members / favorites / popularity	Total users who added the title to any list (Watching, Completed, Dropped, Plan to Watch), users who marked it a favorite, and MAL's own popularity rank (1 = most popular).
season / year / broadcast_day	Release season, release year, and day-of-week of broadcast.
synopsis_clean	The show synopsis after HTML tags, MAL boilerplate, and source attributions were stripped.
balanced_score	The engineered, bias-corrected success metric introduced in this project.

Table 2.1 Data dictionary for key attributes used throughout the analysis.

2.2 Data Preprocessing

2.2.1 Handling Missing Values

To address the issue of incomplete attributes in the raw data, a systematic process for handling missing values was implemented, ultimately resulting in a fully prepared working dataset containing 8,227 records.

During the initial cleaning phase, the “broadcast_day” and “year” fields were directly interpolated by extracting the day of the week and the year of release from the “aired_from” timestamp, as both are deterministic functions of the provided broadcast date. The “licensors” attribute (with a missing value rate of 77%) was excluded because licensing arrangements vary by regional distributor, and this attribute does not affect upstream production attributes as much as major studios or production companies do. Similarly, since the “aired_to” is not essential for evaluating market outcomes, they were excluded. Missing values for the number of episodes were imputed by calculating

the median number of episodes within each anime type (TV, movie, ONA, OVA) such as the median for movie series is 1 episode, while the TV series median is 23 episodes.

The "duration" field requires extra attention. Instead of keeping the free text format (e.g. "24 min per episode"), a regex extracts hour/min/sec values from the text string, which are converted to an integer, "duration_minutes". After checking the parsed values, 456 extremely short (under 1 min) ONA entries were found. These entries almost certainly consist of promotional clips, opening and closing sequences, or commercials, rather than full-fledged short animations. For the "duration_minutes" feature, rather than adopting a single, universal threshold, this approach employs minimum duration thresholds tailored to different type of anime; Movies ≥ 15 minutes, TV series ≥ 3 minutes, ONA ≥ 1 minute, and OVA ≥ 10 minutes. This approach preserves legitimate short-form works within each format (short theatrical movies, short TV series, and short OVAs) while filtering out low-quality promotional content. Combined with the removal of explicit adult content (Rx – hentai), these criteria filtered out invalid entries before the results were modeled.

For the "score" and "scored_by" fields, data missing primarily stems from two causes: first, occasional errors during API scraping; and second, some titles in the MyAnimeList [2] database lack official ratings altogether due to insufficient votes. Because both attributes and the composite "balanced_score" derived from them, rows missing either field were completely removed rather than imputed with fabricated scores.

The "demographics" tag is absent for most entries; this reflects a structural catalog property rather than missing data, as general-audience titles are rarely assigned demographic labels by community contributors. Therefore, missing "demographic" values were preserved as informative categories labeled "Unspecified" rather than imputed artificially.

After dropping the score/scored_by/synopsis-null rows, nulls in producers, and studios contributed less than 7%. Therefore, rows for high visibility titles that fell under one of the criteria established "(member > 10,000 / score > 8.0 / top 1500 popularity)" were all populated and completed with information extracted lookup against Google and Wikipedia. Following that, the missing producers and studios entries were filled with an explicit 'Unknown'/'Other' category. The process resulted in 8,227 fully

completed anime entries, which were utilized for the remaining stages of analysis and modeling.

2.2.2 Data Transformation

Several multi-valued classification fields like genres, themes, studios, producers, and demographics are stored as string representations of Python lists, e.g. ['Action', 'Adventure']. These fields are unquoted using ``ast.literal_eval`` so that individual values can be expanded and encoded for further analysis. The raw synopsis field was also cleaned using regular expressions to strip HTML tags, metadata markers (such as [Written by MAL Rewrite]), and whitespace, generating the 'synopsis_clean' attribute for text mining.

The entity normalization was carried out to handle various differences in names of studios, producers, genres and themes. "difflib.SequenceMatcher" is used to identified variations in studio name due to differences in spelling or punctuation. The same studio name is associated with a single canonical name after manual verification. The normalization process prevents the entity from appearing multiple times and results in a consistent association rule mining process and performance analysis.

2.2.3 Feature Selection and Engineering

The selection of features was carried out to focus on their relevance to the analysis and modeling tasks. The chosen attributes consist of: studio, producer, genre, theme, source, type, demographics, season, year, broadcast day, episodes, duration, synopsis, score, scored_by, rank, popularity, members, and favorites. When attempting prediction, only features that exist prior to or on the release of an anime have been used, excluding features like member, favorites and popularity, to avoid data leakage into the training set.

MyAnimeList [2] does not display a simple arithmetic mean of user ratings. Instead, it publishes a Bayesian-adjusted weighted score, calculated as:

$$\text{Weighted Score} = (v \div (v + m)) \times S + (m \div (v + m)) \times C$$

where S is the average (mean) score given by users who have rated that specific anime, v is the number of valid user votes/scores for that specific title, m is the minimum number of votes required for a title to get a reliable weighted score (historically set to 50), and C is the mean score across the entire MAL database [3]. The weighted score normalizes the influence of titles with a small number of ratings by adjusting their scores toward the overall MAL average. In general, a higher weighted

score represents audience acceptance (perception) of an anime and can therefore act as a quality measurement for audience perception. However, the weighted score is a relatively insufficient for defining anime success because it only measures audience reception quality, not audience presence.

A title may still receive a high weighted score even if their audience may be small, while a popular title may get a less significant score due to ratings from a larger and more diverse audience. Therefore, a high score may not mean high overall success and a low-value doesn't necessarily mean poor performance. To correct for this, a single composite balanced score was engineered for this project to provide a better picture of success for anime,

$$\text{balanced_score} = 0.40 \times \text{score_pct} + 0.25 \times \text{members_pct} + 0.20 \times \text{favorites_pct} + 0.15 \times \text{popularity_pct}$$

Component	What it captures	Transform used
score (40%)	Critical / audience reception	Percentile rank among rated titles
members (25%)	Total reach — everyone who ever added the title to any list	Percentile rank of “log1p(members)”
favorites (20%)	Depth of fan enthusiasm — users who loved it enough to favorite it	Percentile rank of “log1p(favorites)”
popularity (15%)	MAL's own popularity rank (1 = most popular, so inverted)	Percentile rank of “-popularity”

Table 2.2 Composition of the engineered balanced_score metric.

Each component was converted to a 0–1 percentile rank, with “log1p” applied to “members” and “favorites” to reduce their right-skewed distributions. For the final 8,227 titles, balanced_score had a mean of 0.500 and a standard deviation of 0.260, with a strong correlation with raw score (($r = 0.845$)). This indicates that the measure remains broadly consistent with audience reception while also accounting for differences in audience reach.

Title	score	members	popularity	favorites	balanced_score
Zhe Tian Movie: Bei Guan Zhan Wang Teng	7.69	760	16,529	0	0.370
You Shou Yan 3rd Season	7.93	1,131	14,925	8	0.461
Wu Dong Qian Kun	7.92	1,580	13,539	5	0.467

Table 2.3 High raw score but low reach: Balanced Score Reduction

Title	score	members	popularity	favorites	balanced_score
Ousama Game The Animation	5.08	269,007	1,056	522	0.514
Diabolik Lovers	5.18	375,821	738	1,905	0.555
One Punch Man	4.76	447,981	599	7,453	0.568
Boruto: Naruto Next Generations	5.97	936,027	217	7,453	0.646

Table 2.4 Low-to-mid raw score, very high reach: Balanced Score Increase

Tables 2.3 and 2.4 illustrate how the “balanced_score” changes the assessment of anime performance when audience reach is considered alongside rating score. For instance, *Zhe Tian Movie: Bei Guan Zhan Wang Teng* shown in Table 2.3 has a relatively high raw score of **7.69**, but it has a small amount of **members 760**, resulting in a balanced_score of **0.370**. In contrast, *Boruto: Naruto Next Generations* having raw score **5.97** gets an audience members up to **936,027** which brings a balance_score of **0.646**. These comparisons demonstrate that balanced_score considers both audience reception and reach, providing a broader representation of anime performance.

2.3 Data Visualization

Data visualization was applied during all stages of preprocessing to get an idea of the overall shape of the data, reveal remaining quality issues and guide the mining processes at subsequent stages.

2.3.1 Numeric Feature Distribution

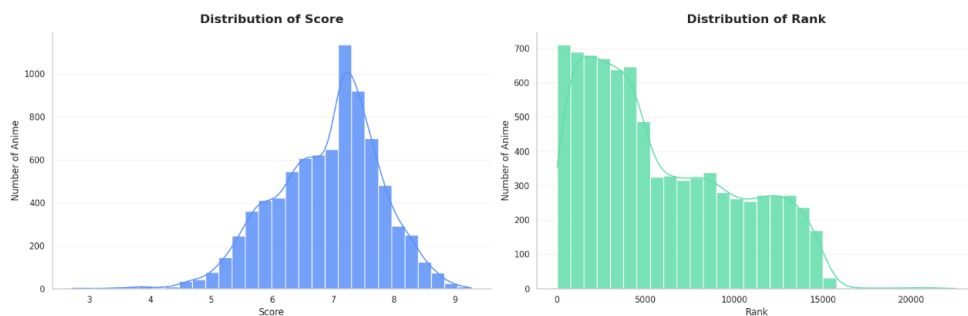


Figure 2.1 Distribution of score and rank

The distribution of users' raw score is around a normal distribution centered between 7.0 and 7.5 indicating a skew in how community members rate their completed titles. The distribution of rankings is strongly focused at the top 5,000 and slowly diminish up to the long tail associated to the low-ranked titles.

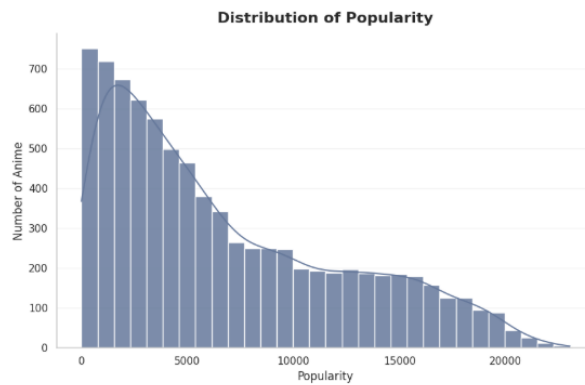


Figure 2.2 Distribution of popularity

The distribution of popularity is heavily concentrated within the top 5,000 titles and gradually diminishes into a long tail associated with less popular anime.

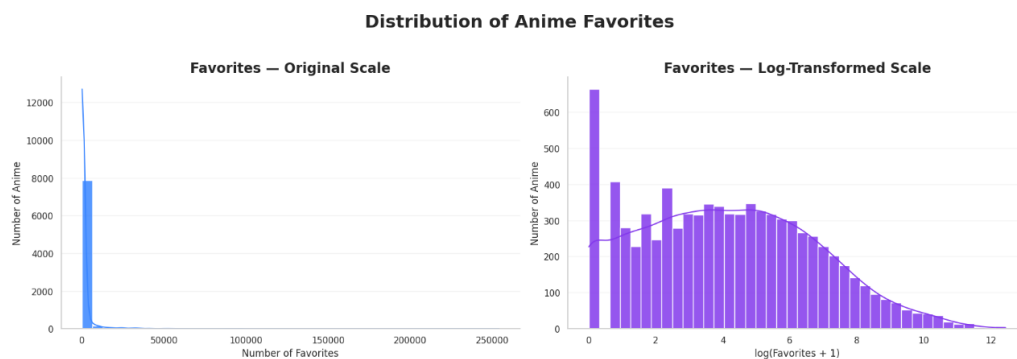
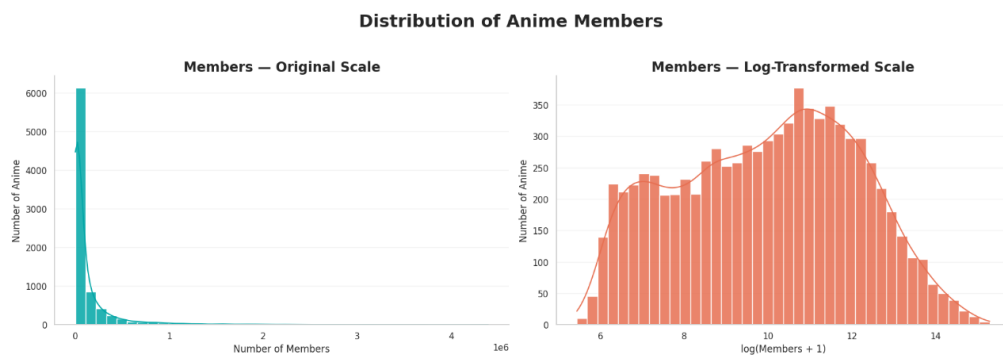
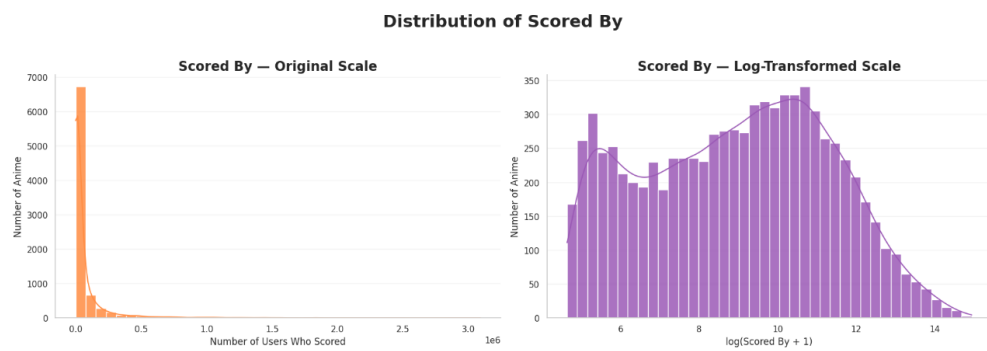


Figure 2.3 Distribution of Engagement signals

In Figure 2.2, three engagement signals (Scored_By, Members, and Favorites) are highly right-skewed on their true scale due to a handful of viral titles; however, after a log transform, a clear engagement funnel appears: Members is the lowest barrier, broadest reach, Scored_by shows active viewer judgment and completion efforts, Favorites feature is a highly exclusive indicator of loyalty, and its pronounced zero peak suggests that such deep emotional attachment remains extremely rare across the catalog.

2.3.2 Average Episode and Runtimes by Media Type

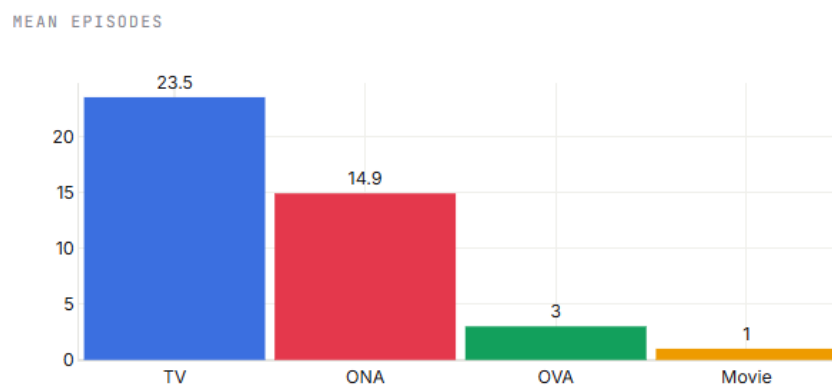


Figure 2.4 Average number of episodes by anime type

Figure 2.3 illustrates that the highest quantity of episodes corresponds to TV shows, whose average series have 23.5 episodes. This reflects typical seasonal broadcast structures. ONAs are second with an average of 14.9 episodes, and OVAs have a drastically lower average number of 3.0 episodes per series, while standalone theatrical Movies remain fixed at 1.0 episode.

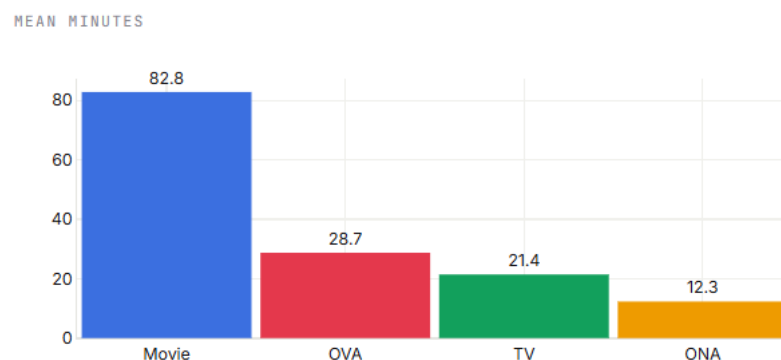


Figure 2.5 Average number of durations by anime type

Figure 2.4 justify that format differences clearly exist in the distributions of duration lengths. The median for Movies is 82.8 mins, for OVAS it is 28.7 mins, TV episodes cluster tightly around 21.4 minutes to match broadcast slots and ONAs average the shortest at 12.3 minutes with broad variance.

2.3.3 Genre Trend Over Time

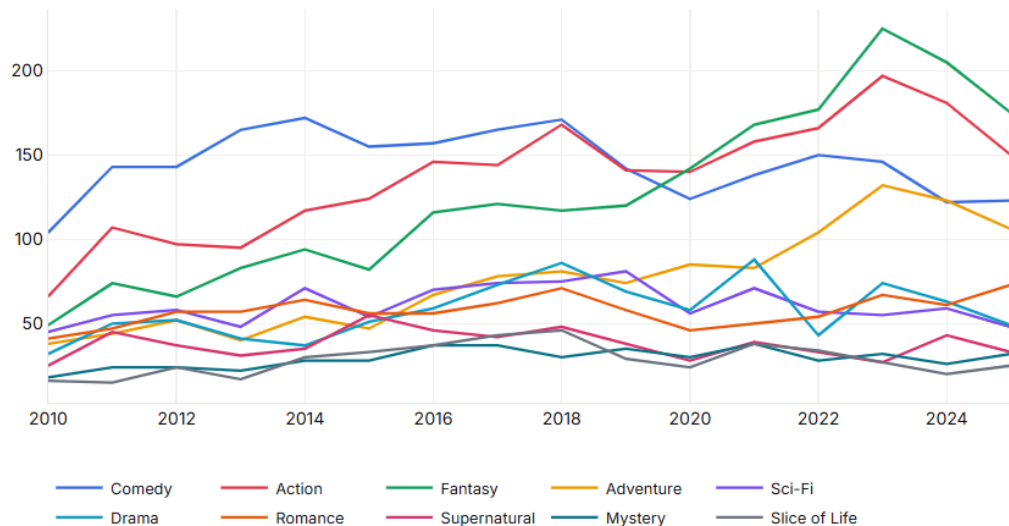


Figure 2.6 Top-10 genre release-count trend, 2010–2025

Tracking annual release numbers by genre from 2010 through 2025, we can see how this market is shifting drastically. Comedy was the primary genre at the end of the last decade, and is steadily dropping as of 2018. Meanwhile, fantasy and action genres are on a sharp rise, now being the two predominant genres, both peaking at 2019, and having at least or close to 200+ releases each. Romance and Sci-Fi have both increased by far more moderate numbers and remain largely flat, while the least yearly fluctuations come from the Slice of Life and Mystery genres.

CHAPTER 3

APPLIED METHODOLOGIES AND MINING

In this chapter, cleaned MyAnimeList [2] dataset was utilized and analyzed using both descriptive and predictive data mining methods to study patterns on production, content, audience reception and commercial success of anime. This descriptive analysis includes association rule mining for relationships between categorical production and content attributes, and text mining for unveiling latent narrative from synopsis. The predictive mining in the study focuses on three analytical tasks: anime recommendation based on content similarity, anime pre-release success classification, and anime demand estimation based on historical temporal patterns.

3.1 Descriptive Mining

The descriptive analysis attempts to answer the following three research questions:

- RQ1: What genre and theme combinations occur most frequently across the anime catalog?
- RQ2: Which producers, studios, and producer–studio collaborations are most strongly associated with anime success?
- RQ3: What narrative patterns can be identified from anime synopsis text beyond the platform’s predefined genres and themes?

To address these research questions, two approaches were applied. The relationships between the various genres, themes, studios, producers and success were explored using an FP-growth association rule mining. A synopsis text mining was applied to identify groups of anime with similar narrative characteristics based on their plot descriptions.

3.1.1 Association Rule Mining

In order to answer RQ1 and RQ2, meaningful relationships between content & production attributes within the anime dataset were analyzed by using FP-Growth [4].

Genre ↔ Theme Co-occurrence

The genre-theme analysis seeks to identify common co-occurrence across content categories independently of the success categories. Results here identify highly specific associations such as Team Sports → Sports (100% confidence, lift = 22.62) and Detective → Mystery (99.40% confidence, lift = 12.73), which confirms the

reliability of the catalog annotations. The more informative associations show the various elements of content that form specific types of narratives. Fantasy, Action, Historical, Martial Arts were strongly associated with 72.70% confidence, while Sci-Fi and Military, and Mecha are also strongly associated with a lift of 9.79. Similarly, Psychological and Suspense had high lift of 16.52. These associations show important mixes of genres and themes reflecting commonly observed recurring patterns of storytelling in anime, giving a clearer overview of the different content pieces being put together within the catalog.

#	Association	Support	Confidence	Lift
1	Theme_Team Sports → Genre_Sports	1.90%	100.00%	22.62
2	Genre_Adventure + Theme_Martial Arts → Genre_Fantasy + Theme_Historical + Genre_Action	2.17%	62.41%	16.62
3	Theme_Psychological → Genre_Suspense	2.38%	66.32%	16.52
4	Theme_Idols (Female) → Theme_Music	1.54%	83.33%	14.93
5	Genre_Fantasy + Theme_Historical + Genre_Action → Theme_Martial Arts	2.73%	72.70%	13.29
6	Theme_Detective → Genre_Mystery	2.05%	99.40%	12.73
7	Theme_Harem → Genre_Ecchi	1.88%	51.70%	10.82
8	Genre_Sci-Fi + Theme_Military → Theme_Mecha	1.67%	57.94%	9.79

Figure 3.1 Genre ↔ Theme Co-occurrence

Producer Strategy and Commercial Outcomes

The producers were also identified to be consistently highly performing when together with specific genres or themes.

- Aniplex showed strong associations with high success in Supernatural, Fantasy, Action, Romance, and School content.
- Shueisha was primarily associated with high success in Action and School theme anime.
- Dentsu showed a strong association between Action productions and high success.

These results lead one to conclude that producers may indeed not operate independently of content type; particular producers are shown repeatedly to appear within categories where they seem to have a track record of success.

No.	Association Rule	Support	Confidence	Lift
1	Producer_Aniplex + Genre_Supernatural → Balanced_High	1.16%	75.20%	3.01
2	Genre_Fantasy + Producer_Aniplex + Genre_Action → Balanced_High	1.00%	75.00%	3.00
3	Producer_Aniplex + Genre_Romance → Balanced_High	1.10%	70.63%	2.82
4	Producer_Aniplex + Genre_Action → Balanced_High	2.10%	67.19%	2.69
5	Genre_Action + Producer_Shueisha → Balanced_High	1.77%	67.14%	2.68
6	Theme_School + Producer_Aniplex → Balanced_High	1.07%	64.93%	2.60
7	Theme_School + Producer_Shueisha → Balanced_High	1.12%	63.19%	2.53
8	Producer_Dentsu + Genre_Action → Balanced_High	1.42%	62.84%	2.51

Figure 3.2 Top producer + genre association rules for the Balanced_High outcome

Studio and Success Associations

Likewise, at a studio level, similar relationships could be observed between High Success and individual genre areas.

- For Action, a high success connection was observed with both A-1 Pictures and Bones. They had confidence above 65% with lift values over 2.6.
- Madhouse was strongly associated with high success in Drama, with a lift of 2.26 and confidence of 56.58%.
- J.C. Staff showed a positive association with Comedy, with a lift of 1.78 and confidence of 44.44%.
- For Toei Animation, consistently high success appeared with Adventure and Action.

What the findings show is that successful studios are likely to be found repeatedly in certain genre areas rather than performing in a uniform way across all kinds of anime.

	antecedents	consequents	support	confidence	lift
0	(Genre_Action, Studio_A-1 Pictures)	(Balanced_High)	0.006199	0.689189	2.756422
1	(Genre_Action, Studio_Bones)	(Balanced_High)	0.005348	0.656716	2.626546
2	(Genre_Drama, Studio_Madhouse)	(Balanced_High)	0.005227	0.565789	2.262883
3	(Genre_Comedy, Studio_J.C.Staff)	(Balanced_High)	0.005834	0.444444	1.777562
4	(Studio_Toei Animation, Genre_Adventure)	(Balanced_High)	0.005105	0.375000	1.499818
5	(Genre_Action, Studio_Toei Animation)	(Balanced_High)	0.005470	0.304054	1.216068

Figure 3.3 Top studios + genre association rules for the Balanced_High outcome

3.1.2 Producer-Studio Collaboration Patterns

To further examine RQ2, producer-studio partnerships with at least 5 shared productions were explored using average `balance_score`.

- Studio Bind: Collaborations with Frontier Works, TOHO Animation, Hakuhodo DY, and KADOKAWA achieved average scores of 0.935–0.961, largely associated with the *Mushoku Tensei* franchise.
- David Production and Medicos Entertainment; achieved 0.956 by average of 8 projects, mostly relevant with the *JoJo's Bizarre Adventure* series.
- Studio Ghibli and Dentsu; achieved 0.941 with 6 movies of theatrical production series by average.
- Clover Works: Collaboration with Toho Animation, Shogakukan-Shueisha, TV Tokyo, resulted in average 0.931-0.939, largely related with titles like the *Spy-X Family* franchise.

Therefore, title-level inspection reveals that the most successful collaborations between producer and studio were mainly sustained through re-workings of existing series and sequels, rather than isolated productions.

3.1.3 Synopsis Text Mining and Narrative Clustering

Research Question 3 (RQ3) is approached by inspecting synopses of anime to examine narrative similarity, without using predefined genre labels. This study applied 6798 anime records with usable synopsis text, which were obtained by keeping records whose synopsis was at least 150 in length. Synopses of anime are stripped of English stopwords, converted to TF-IDF representations [5]. However, the produced TF-IDF matrix is very high dimensional and sparse, so applied TruncatedSVD to get low-dimensional 30 features latent representation of each synopsis. Before clustering, extracted vectors were L2-normalized. Following, performed K-Means clustering [4] among possible values for parameter *k* ranging from 8 to 40. Optimal value achieved for *k*=32 that produce 0.1488 silhouette score. While this silhouette score is poor, this is quite to be expected for narrative material, where anime could easily and often share common background setting, themes and narrative elements

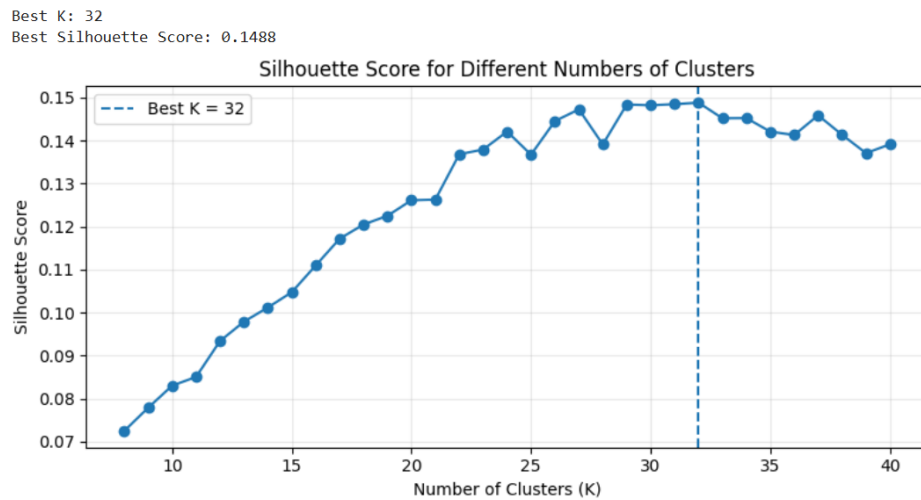


Figure 3.4 Silhouette score across candidate cluster counts k = 8-40

In order to examine whether the narrative clusters were significantly associated with genre, the cluster membership was cross tabulated with MAL's genre labels and evaluated with a chi-square test for independence [4]. The resulting value was statistically significant ($\chi^2 = 10,372$, $p < .001$). Therefore, the clusters are indeed related to genre labels. However, because the value of Cramér's V [4] is only 0.180, there is only a weak association and suggests that the clusters capture narrative patterns that are related to genres, but not simply reproduce predefined genre classifications.

Identified Narrative Patterns

Inspecting the dominant TF-IDF keywords and genre distributions demonstrated that these 32 clusters identified distinct and readily interpretable narrative prototype patterns.

- Franchise & mascot narratives: Clusters 0, 16 and 20 represented recognizable franchise-based narratives of Pokemon, One Piece and Doraemon.
- Fantasy & sci-fi narratives: Cluster 1 represents high-fantasy narrative setting, while Cluster 19 deals with cultivation/xianxia narrative. Cluster 18 focuses on science-fiction and space conflict narratives and Cluster 27 focuses on demon hero narratives.
- School, club & romance narratives: Clusters 9, 8 and 31 correspond to the more general high school setting, with a focus on school club, or student council activity respectively. Cluster 21 identifies romance and relationship centric narratives.
- Specialized narrative patterns: Clusters 13, 17, 22, and 25 capture sport, detective, idol/music, and martial arts narratives respectively.

In conclusion, these clusters exhibit how plot summaries reveal narrative structures, settings, franchises and recurring story patterns not available by only looking at the pre-assigned genre and tag information and provide an alternative point of view to understanding an anime content.

3.1 Predictive Mining

Predictive mining was carried out to build predictive models useful for three forward-looking analytical tasks: anime recommendation based on content features, classifying success of anime prior to its release, and forecasting audience-demand for specific genres.

3.2.1 Content-Based Hybrid Anime Recommendation

A hybrid content-based recommender was developed aiming at finding anime with similar story and content characteristics. Story similarity was achieved by taking the cosine similarity of 100 components TruncatedSVD embeddings generated from TF-IDF synopsis vectors and genre/theme similarity was obtained from multi-hot encoded genre/theme vectors. Both of the similarity measures were min-max scaled and fused by a weighted sum:

$$\text{hybrid_similarity} = 0.6 * \text{similarity_story} + 0.4 * \text{similarity_genre-theme}$$

Story similarity received a greater weight to ensure the matches prioritized narrative components, while genre and theme information provided an additional content match. Fuzzy title matching was performed across English, Japanese and synonymous title forms to improve query finding (e.g., resolving *進撃の巨人* to *Shingeki no Kyojin*).

3.2.2 Anime Success Classification

The success classification task aims at classifying if a given anime is a commercial success based on available information pre-release. It removes post-release variables such as members, scores, popularity, favorites, and balanced_score that post-release variables such as member counts, scores, popularity, and favorites were excluded to prevent data leakage.

Modeling

Four machine learning algorithms, Logistics Regression, Decision Tree, Random Forest [7] , and XGBoost [6] were developed for the success classification.

The models were trained with a combination of pre-release production, content and narrative features in order to classify an anime was likely to achieve high balanced performance or not.

Applied Methodologies

In this study, Logistic Regression was selected as a linear baseline, whereas Decision Tree and Random Forest were elected to exploit non-linearity of the relationships. XGBoost, a gradient boosting model, was used to model complex interactions among features.

Since the “success” class -1 represented approximately one-quarter of the dataset, two approaches to class imbalance were explored: class weighting and SMOTE (Synthetic Minority Over-sampling Technique). These approaches were compared using the four classification algorithms to select the best approach to this class imbalance problem.

Model Setup

The modeling dataset uses the 6659 completed anime titles that was released from 2010. To reflect a real-world prediction task and maintaining the time dependencies, the data were separated into training set, validation set and test set based on the chronological year that the title was released

- Training Set (2010 - 2022): 5165 titles (77.6% of data)
- Validation Set (2023): 480 titles (7.2% of data)
- Test Set (2024 - 2025): 1014 titles (15.2% of data)

The `is_success_b` feature was defined as “1” if `balanced_score` were above or equal to 0.6946 which is the 75th percentile in the training set's `balanced_score` distribution, 0 otherwise. The "success" label is 25% in training, 25% in validation, and 23.4% in test.

A custom FeatureBuilder produces a set of 108-dimensional features for each anime based on the pre-release information. The features included categories such as the genre, themes, studios, producers, demographics, source, type, age rating, and production metadata such as episodes, duration, release season. Also, it includes the historic performance of related studio and source material and a 20-dimensions SVD representation for the anime synopsis.

Baseline Model

Logistic Regression was chosen as the baseline model because it provides a straightforward, linear benchmark to which the tree-based models can be compared. It serves as a benchmark for determining if non-linear methods are effective at improving classification results over a simple linear approach.

Imbalance Handling and Time-Series Cross Validation

Two imbalance-handling approaches were studied: class weighting and SMOTE. Under the class weighting strategy, “class_weight='balanced'” was used for Logistic Regression, Decision Tree, and Random Forest models to give more importance to the minority class during the training process of those models. With regard to XGBoost, the problem of imbalance was dealt with through the use of scale_pos_weight by taking the ratio of non-success to success training examples to produce a value of 2.9977. Therefore, misclassification of success anime carried about three times more weight than the non-success anime.

With the SMOTE technique, the generation of synthetic data for the minority success class was performed using the training dataset before training the models. The training dataset, which had an imbalance of classes, was augmented to 7,746 samples having a balance of 50:50 of both classes. The validation and test sets were never oversampled.

The approach that produced better-performing models was then used for further validation. As the anime data is a temporal sequence, the next validation step was to use an “expanding-window time-series cross-validation” to assess model stability across different historical periods. the models across time. Each validation fold increased the training period sequentially: 2010-2018 → 2019, 2010-2019 → 2020, 2010-2020 → 2021, 2010-2021 → 2022.

The results from cross-validation were used to choose final classification model based on its predictive performance and stability over different time periods.

Proposed Model and Optimization

XGBoost was chosen as the proposed model because it has the potential to capture non-linearities and interactions between the tabular and text features. The selected model was then optimized by adjusting the classification threshold on the

validation set for the maximum F1 score and thereafter this threshold was fixed for final evaluation on the hold-out test set. SHAP was also used for model interpretation and feature importance determination [8].

3.2.3 Genre Trend and Audience-Demand Forecasting

The third predictive problem deals with future production volume of different genres and the audience demand of each genre, using the historical data range from 2010 to 2025. The prediction was carried out by developing two complimentary forecasts: one predicting the volume of anime releases per year, which focuses on the top six highest-volume genres: Comedy, Action, Fantasy, Adventure, Sci-Fi and Drama; the second one forecasting the demand for six selected genres of audience interest based on audience-engagement volatility: Suspense, Romance, Ecchi, Girls Love, Mystery and Supernatural using median MAL member counts as the demand measure. In both forecasts, the 2010-2022 data were first used to forecast 2023-2025 for backtesting, then test against actual data and evaluated in measures of MAE, RMSE and MAPE. Finally, each model was retrained with all the data from 2010-2025 for forecasting of 2026 to 2028.

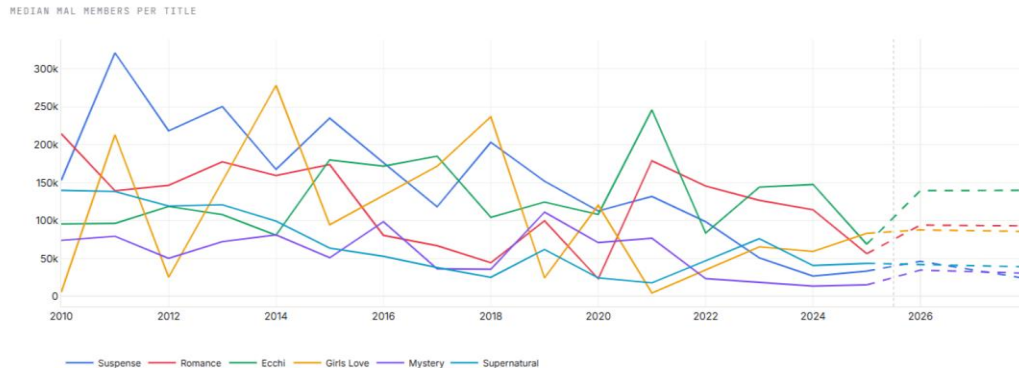


Figure 3.5 Genre audience-demand forecast

Genre	2026 forecast	2027 forecast	2028 forecast	Trend
Ecchi	139,427	139,630	139,792	Slight increase
Romance	93,981	93,401	92,934	Slight decrease
Girls Love	87,589	86,579	85,574	Decrease
Supernatural	41,848	40,279	38,923	Decrease
Mystery	34,624	32,553	30,492	Decrease
Suspense	46,250	34,719	23,245	Sharp decrease

Table 3.1 2026–2028 genre audience-demand forecast.

According to the Holt-Winters forecast [9], Ecchi will continue to have high projected audience demand while slowly increase throughout 2028. Romance and Girls Love gradually decreases while Supernatural and Mystery decrease more substantially. Suspense has the largest predicted decrease going from 46,250 median members in 2026 down to 23,245 members in 2028.

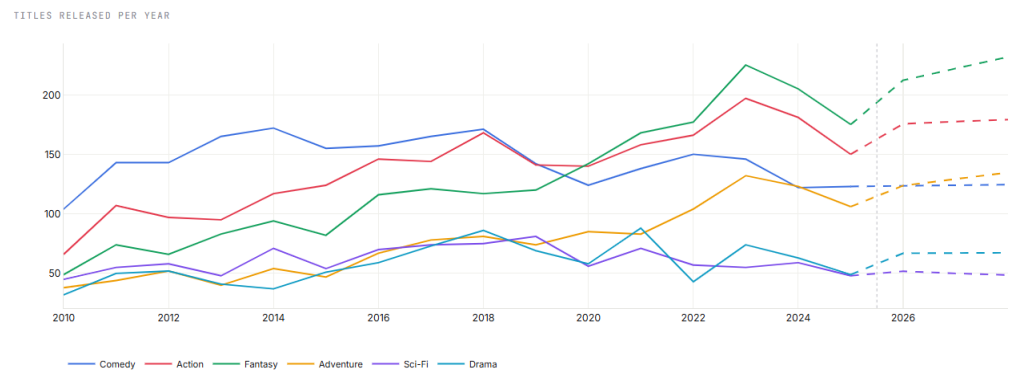


Figure 3.6 Genre release-volume forecast

Genre	2026 forecast	2027 forecast	2028 forecast
Fantasy	212	222	232
Action	176	178	179
Adventure	124	129	135
Comedy	124	124	124
Drama	67	67	67
Sci-Fi	52	50	49

Table 3.2 2026–2028 genre release-volume forecast

In here, the forecast indicates that release volumes for Fantasy, Action and Adventure expected to increase until 2028. Comedy and Drama are expected to remain stable, whilst Sci-Fi slightly decreases.

CHAPTER 4

EVALUATION

This chapter discusses the results and evaluation of the predictive mining tasks performed in this study. Evaluation includes assessment of the hybrid anime recommendation system, anime success prediction, and genre trends and demands forecasting. Performance of the models is evaluated using quantitative metrics along with qualitative analysis where appropriate.

4.1 Anime Recommendation System Evaluation

The hybrid recommendation system was evaluated using similarity-based measures and qualitative analysis of the generated recommendations. Since there was no available labeled set of the correct recommendations, the evaluation sought to find out whether the recommended titles had significantly higher hybrid similarity than random titles while at the same time looking into whether the retrieved titles had significant storyline, genres, themes, and franchise connections.

A set of 500 anime titles was chosen for testing. The average similarity of the top 10 recommendations generated using the hybrid approach was 0.656 against to 0.129 for randomly generated pairs. Thus, the similarity of the recommendations produced using the hybrid recommendation approach is 5.07 times greater than the similarity score of the randomly selected pairs.

The qualitative analysis also showed the relevant franchise and storyline connections. For instance, *Fullmetal Alchemist: Brotherhood* suggested titles associated with the same franchise and dark fantasy/military narratives, while *Death Note* retrieved complex psychological and supernatural thrillers. The results give evidence of the role of storyline information in the recommendations.

In conclusion, it can be stated that the recommender system creates reasonable content-based recommendations on the basis of content similarity in the form of storyline, genre and themes. However, the evaluation currently performed does not prove the optimality of this hybrid recommendation system. Further evaluation by users or experts remains necessary to evaluate the relevancy and practical quality of the recommendations.

4.2 Anime Success Classification Performance Evaluation

The evaluation of anime success tested four different models through two imbalance resolution methods, which were class weight adjustment and SMOTE. From the experimental outcomes, the method that performed better was used in expanded window time series cross validation.

4.2.1 Comparison of Experimental Results

Two imbalance-handling approaches were explored to address the unequal distribution across the successful and non-successful anime. The class weighting approach was compared against a SMOTE-based approach that balance the training samples. All the four models were evaluated for accuracy, precision, recall, F1-score, and ROC-AUC using each approach.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7130	0.4366	0.7848	0.5611	0.8243
Random Forest	0.7101	0.4379	0.8481	0.5776	0.8422
Decision Tree	0.6923	0.4161	0.7848	0.5439	0.7622
XGBoost	0.7091	0.4350	0.8186	0.5681	0.8453

Table 4.1 Performance of Models under Class Weighting

From class weighting, all four models had high recall meaning the models managed to correctly recall most of the successful anime. XGBoost had the highest ROC-AUC **0.8453**, while Random Forest had the highest recall and F1-score.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7130	0.4366	0.7848	0.5611	0.8223
Random Forest	0.6972	0.4249	0.8354	0.5633	0.8336
Decision Tree	0.6696	0.3975	0.8017	0.5315	0.7653
XGBoost	0.7564	0.4858	0.7215	0.5806	0.8310

Table 4.2 Performance of Models under SMOTE

Although SMOTE generated synthetic minority class data to balance the training data, this doesn't mean better prediction results for the 2024-2025 test data. The ROC-

AUC score for XGBoost, Random Forest, and logistic regression was poor, although the performance of Decision Tree was marginally better.

Thus, the class-weighting approach was finally selected to validate through time series validation as this gave the best ROC-AUC results for the four models, especially XGBoost.

4.2.2 ROC and AUC

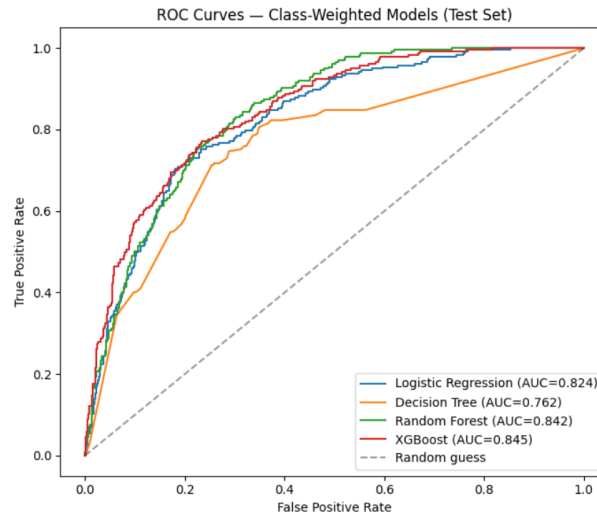


Figure 4.1 ROC- Class-weighted Models (Test Set)

The ROC curves show that the class-weighted models have significantly different ability to discriminate the classes. The Decision Tree (0.762) performs worse than all other models over most of the range of false-positives. XGBoost (0.845) and Random Forest (0.842) have the strongest abilities to discriminate the classes over the full range of false-positives and are closely comparable. The logistic regression is showing good class separation but slightly less than the other two ensemble models.

4.2.3 Time-Series Cross-Validation and Threshold Optimization

The selected class-weighting approach was tested using expanding-window time-series cross-validation to accommodate temporal changes in anime production and patterns. Four chronological validation folds were employed, with the training period gradually extended from 2010–2018 to 2010–2021.

Model	Accuracy (Mean $\pm$ SD)	Precision (Mean $\pm$ SD)	Recall (Mean $\pm$ SD)	F1 (Mean $\pm$ SD)	ROC-AUC (Mean $\pm$ SD)
Logistic Regression	0.8282 $\pm$ 0.0183	0.6228 $\pm$ 0.0470	0.8035 $\pm$ 0.0735	0.7003 $\pm$ 0.0456	0.8965 $\pm$ 0.0226
Random Forest	0.8277 $\pm$ 0.0121	0.6331 $\pm$ 0.0471	0.7605 $\pm$ 0.0447	0.6891 $\pm$ 0.0205	0.8990 $\pm$ 0.0141
Decision Tree	0.7756 $\pm$ 0.0244	0.5413 $\pm$ 0.0493	0.7280 $\pm$ 0.0426	0.6199 $\pm$ 0.0399	0.7912 $\pm$ 0.0303
XGBoost	0.8357 $\pm$ 0.0089	0.6468 $\pm$ 0.0463	0.7817 $\pm$ 0.0909	0.7038 $\pm$ 0.0247	0.9074 $\pm$ 0.0173

Table 4.3. Expanding-Window Time-Series Cross-Validation Results

XGBoost had the highest mean accuracy, precision, F1-score, and ROC-AUC for the four expanding-window folds. It also had the lowest accuracy and F1-score standard deviation which suggested that it performed best over different periods of history. Thus, XGBoost is utilized as the final classification model.

Then the classifier threshold was optimized on the 2023 validation set. Various thresholds, from 0.30 to 0.80, were tested and 0.63 threshold gave in the highest validation F1-score. Therefore, 0.63 threshold was used in the final testing of the hold-out test set.

Threshold	Accuracy	Precision	Recall	F1-score
0.63	0.8417	0.6346	0.8390	0.7226

Table 4.4 Validation Threshold Optimization

4.2.4 Final XGBoost Evaluation and Confusion Matrix

The final XGBoost model was evaluated on the 2024-2025 hold-out test set with the best threshold at 0.63.

Threshold	Accuracy	Precision	Recall	F1-score	ROC-AUC
0.63	0.7663	0.5000	0.7722	0.6070	0.8453

Table 4.5. Final XGBoost Performance

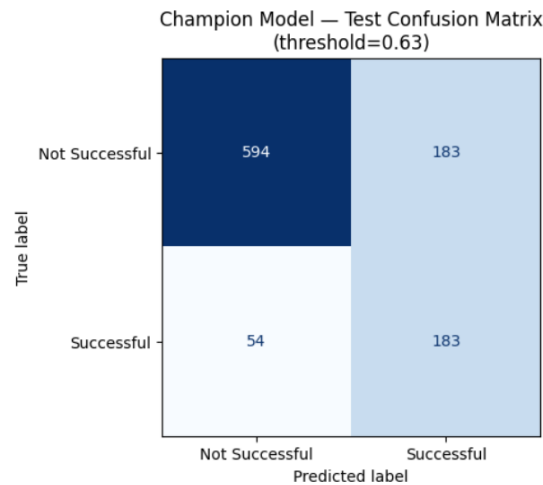


Figure 4.2 XGBoost - Test Confusion Matrix

The confusion matrix indicates that the model could identify 183 successful anime out of 237. However, 183 failed anime were identified as successful anime, which resulted in the precision of 0.5000. In other words, the model was effective for capturing a wider range of potential successful anime, but not for identifying all positive cases.

4.2.5 Findings

Based on the experiments, class weighting approach outperformed SMOTE approach, although both improved minority-class detection. SMOTE equalized the training dataset to be equal by class, but did not consistently outperform the original dataset test performance distribution.

XGBoost received the best time-series cross-validation performance, with a mean ROC-AUC of 0.9074 and mean F1-score of 0.7038. After adjusting the optimal threshold for the model, the final model resulted in 0.8453 ROC-AUC, 0.7663 accuracy, 0.5000 precision, 0.7722 recall, and 0.6070 F1-score on the hold-out test set. These results show that the model has a good ability to make predictions. However, the moderate precision means that distinguishing between successful and unsuccessful anime remains somewhat challenging.

4.3 Forecasting Evaluation

To evaluate forecast accuracy, both Holt-Winters forecasting tasks were backtested using Mean Squared Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These measures were computed based on

the backtest of the 2023-2025 period and lower values indicate better forecasting accuracy.

4.3.1 Genre Release-Volume Forecasting

The table below summarizes the backtesting results for the release-volume forecasts on the selected genres.

Genre	MAE	RMSE	MAPE	Accuracy interpretation
Drama	11	14	21%	Weak
Sci-fi	16	17	31%	Very poor
Adventure	18	22	14%	Relatively good
Comedy	22	25	18%	Relatively good
Action	22	24	12%	Best
Fantasy	30	33	14%	Relatively good

Table 4.6 Backtest Results for Genre Release-Volume Forecasting

The release-volume backtesting results indicate that most genres had reasonably reliable forecast accuracy. Action had the lowest MAPE (12%), followed by Adventure and Fantasy (14%) and Sci-Fi had the highest MAPE of 31%. While Drama had the lowest MAE and RMSE, its errors are quite large in relative to the smaller number of releases. Overall, Action, Adventure and Fantasy had the best forecast performances according to the percentage error.

4.3.2 Genre Audience-Demand Forecasting

The table below summarizes the backtesting results for the audience-demand forecasts on the selected genres.

Genre	MAE	RMSE	MAPE	Accuracy interpretation
Romance	29,596	30,321	36%	Best
Ecchi	30,083	47,662	42%	Relatively good
Supernatural	42,351	44,284	79%	Weak
Mystery	44,929	44,971	291%	Very poor
Girls Love	47,681	48,735	72%	Weak
Suspense	58,949	59,402	175%	Very poor

Table 4.7 Backtest Results for Genre Audience-Demand Forecasting

The audience-demand forecasts varied by genre. Romance performed the best with a MAPE of 36%, and Ecchi was not far behind at 42%. The errors increased for Supernatural (79%) and Girls Love (72%) and then increased dramatically for Suspense and Mystery (175% and 291%, respectively). These results suggest that it is more difficult to forecast audience demand where demand is low or highly unstable.

In contrast, these results suggest that the audience-demand forecasting appears to be less trustworthy than the release-volume forecasting, especially when used to predict genres that have more unstable audience demand historically. Therefore, the Holt-Winters forecast for highly fluctuating genres should be interpreted with greater caution.

CHAPTER 5

CONCLUSION

This project proposes an end-to-end **Anime Industry Analytics System** that leverages MyAnimeList [2] data collected through the Jikan API [1] to generate informative insights into anime production, content, audience engagement and performance. The system integrates data mining, machine learning, text analysis, recommendation, prediction and forecasting to analyze different aspects of the anime industry. These analytical components are integrated into an interactive web-based platform to demonstrate how data driven analysis can transform historical anime information into a decision-support system for understanding industry trends, assessing potential performance, and exploring the future developments in the anime industry.

5.1 Advantages of using this project

This project has several advantages. It consists of:

- ❖ **Data-Driven Decision Making:** The analytical results of production trends, audience engagement, relationship between contents and potential success provides supporting evidence for decision making.
- ❖ **Comprehensive Success Measurement:** The balanced score considers scores, audience reach, favorites, and popularity instead of relying only on audience raw ratings. This provides a broader measurement of anime success.
- ❖ **Pre-Release Success Prediction:** The classification model uses information that was available before the release to predict whether an anime is likely to reach a high level of success, offering useful support in early-stage analysis.
- ❖ **Interpretable Prediction:** The integration of the SHAP analysis demonstrates which features contribute to individual predictions, making the classification results more interpretable.
- ❖ **Integrated Recommendation and Forecasting:** The system integrates storyline-based recommendations and forecasting at the genre level, enabling analysis of various aspects of the anime industry.
- ❖ **User-Friendly Web Interface:** The web user interface is developed to enable the users to interactively browse datasets, analytical results, recommendations, predictions, and forecasts.

5.2 Disadvantages of using this project/ Limitation

Despite its advantages, the system has several limitations including:

- ❖ **Lack of Financial Data:** The dataset does not include budgets, revenue, or marketing costs, so the success measure is based on audience response only rather than financial performance.
- ❖ **Limited Source Material and Franchise Data:** Data on source material popularity, adaptations, or direct franchise connections is not available. Therefore, the influence of the popularity of source material and previous season/franchise entries on anime success are not directly measured.
- ❖ **Limited Recommender Evaluation:** The recommendation engine is evaluated with content similarity rather than user-based accuracy measures or expert evaluation as no data from user preference or interaction is available.
- ❖ **Moderate Classification Precision:** The classifier performed well discriminatively and achieved a reasonable recall, but its moderate precision shows that predicting success pre-release remains challenging using only metadata from the platform.
- ❖ **Dependence on Platform Metadata:** Findings are largely based on MyAnimeList [2] data and may be affected by its coverage, missing data and differences from the broader anime market.

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